



HOSPITAL ADMIN MODULE

Complete Hospital Operating & Administration PRD

Document	Version	Scope	Primary Role
Qlyno Hospital Admin	1.0	Hospital + branches + departments	Hospital Admin
Foundation	HMS + Qlyno ecosystem	OPD • IPD • Emergency • OT • Diagnostics • Pharmacy	Role-based
Special focus	Emergency + network + governance	SOS • ambulance • surgeon • vendor • verification	Human-in-control

Module definition

Hospital Admin is the operational command layer for one hospital organization. It can configure and manage departments, doctors, nurses, Nurse Stations, receptionists, support staff, patients, appointments, OPD/IPD, wards, beds, emergency, OT, diagnostics, pharmacy, billing, inventory, assets, vendors, surgeon requests, ambulance operations, reports, compliance and security.

Admin power - with a critical boundary

Hospital Admin may perform or access operational functions available to other hospital roles when explicitly permitted, including acting on behalf of a staff role for continuity. This must not turn the admin into a clinical decision-maker: diagnosis, prescribing, surgical decisions and other licensed clinical decisions remain attributable to the appropriate clinician.

1. Product Role & Purpose

The supplied HMS Hospital Admin documentation defines the Hospital Admin as the role responsible for day-to-day operations of an individual hospital, including departments, doctors, staff, patients, appointments, billing, inventory, reports and administrative activities. The expanded Qlyno model treats the hospital as an active organization inside a connected healthcare ecosystem rather than an isolated HMS.

Area	Hospital Admin responsibility
Organization	Hospital identity, branches/locations, profile, services, operating rules and public presence.
People	Create/manage doctors, nurses, Nurse Stations, receptionists and support staff.
Clinical operations	Coordinate OPD/IPD, wards, beds, emergency, OT and clinical workflows without replacing clinicians.
Network	Request surgeons, equipment/products and other vendor requirements.
Emergency	Receive and coordinate SOS alerts, pre-arrival emergency workflow and ambulance dispatch.
Patient continuity	Find the existing Qlyno Patient ID and connect hospital treatment to the patient's longitudinal journey.
Governance	Verification, access control, audit, compliance, quality and security.
Performance	Operational, financial, departmental, patient-flow and workforce analytics.

Source alignment

The original HMS Hospital Admin already includes dashboard, departments, doctors, nurses/staff, reception, patients, appointments, OPD, IPD, wards, beds, emergency and OT. The HMS foundation also includes pharmacy, lab, radiology, billing, insurance/TPA, inventory, assets, ambulance, shifts, attendance, notifications, reports, audit logs, documents, communication, quality/compliance, settings, permissions and global search.

2. Hospital Operating Model

Entity	Relationship to Hospital	Control
Hospital	Top-level healthcare organization	Hospital Admin + platform governance
Branch / Location	Optional physical hospital site	Location-scoped access
Department	OPD/IPD/ICU/Emergency/OT/etc.	Department head + Admin
Doctor	Provider affiliated with hospital	Professional verification + clinical role
Nurse Station	Nursing operational unit	Station Lead / Admin
Nurse	Care staff attached to station/unit	Nurse role + scope
Receptionist	Front-desk operational staff	Desk/department scope
Support Staff	Attendant, housekeeping, assistants etc.	Task-based access
Surgeon	Internal or requested external specialist	Case/request-specific access
Vendor	External supplier/network participant	Request/quote/order scope

Patient	Persistent Qlyno identity	Hospital relationship + consent
Ambulance	Hospital transport resource	Availability + dispatch status

3. Hospital Admin Dashboard

Dashboard block	What Admin sees
Hospital Today	Appointments, registrations, admissions, discharges, active emergency cases, procedures and major alerts.
Patient Flow	OPD queue, waiting patients, IPD census, transfers, discharge blockers and emergency arrivals.
Bed Occupancy	Available/occupied/blocked/cleaning beds by ward, ICU/NICU/private/general and other configured units.
Staffing	Doctors, nurses, receptionists and support staff on duty, absent, late, on leave and unassigned.
Emergency	Active SOS alerts, incoming emergency cases, ambulance status, emergency capacity and response state.
OT & Surgery	Today's procedures, OT utilization, surgeon status, readiness blockers and vendor dependencies.
Diagnostics	Pending lab/radiology orders, critical results and delayed reports.
Pharmacy	Pending prescriptions, stock alerts, critical medicine availability and fulfillment exceptions.
Finance	Today's collection, outstanding, refunds, insurance/TPA and reconciliation exceptions.
Procurement	Open vendor requests, quotes awaiting review, purchase orders and delayed deliveries.
Alerts	Security events, access anomalies, overdue tasks, staffing gaps and operational blockers.
Quick Actions	Add doctor, add staff, create department, allocate bed, emergency action, surgeon request, vendor request, report.

4. Doctor Management

Capability	Requirement
Create doctor	Create/invite doctor profile and link to hospital, branch and department.
Professional profile	Specialty, sub-specialty, qualifications, registration details, experience, languages, services and consultation settings.
Verification	Collect required documents and submit for verification; hospital verification does not replace platform-level verification.
Schedule	Configure doctor availability, duty schedule, leave, consultation hours and emergency/on-call status.
Department	Assign department, role, privileges and locations.
Fees/services	Configure hospital-specific consultation/service rules where permitted.
Performance	Appointments, consultation volume, wait time, patient outcomes/operational KPIs where configured.
Public profile	A verified doctor affiliated with the hospital can appear in Qlyno search subject to publication rules.
Deactivate	Suspend/remove hospital affiliation without deleting historical clinical records.

5. Nurse, Nurse Station & Support Staff Management

The Hospital Admin owns the hospital-wide nursing workforce configuration. The Nurse Station then manages day-to-day nursing operations.

Function	Hospital Admin	Nurse Station
Create nurse	Yes	Yes within assigned station
Assign department/station	Yes	Within permitted scope
Create support staff	Yes	Yes for configured station staff
Shift templates	Hospital-wide rules	Create/manage station roster
Roster	View/control hospital-wide	Manage assigned roster
Shift change	Yes	Yes within scope
Patient assignment	Operational override	Primary operational workflow
Handover	Monitor	Create/receive/manage
Staff permissions	Yes	Only delegated permissions
Audit	Full hospital scope	Station scope

The previously defined Nurse Station module supports multiple nurses, support staff, shift templates, rotations, roster changes, patient/task assignment, handovers and doctor coordination. **Hospital Admin remains the higher-level organization owner.**

6. Reception Management

Capability	Hospital Admin requirement
Multiple receptionists	Create multiple receptionists and assign them to desks, branches, departments or workflows.
Front desk	Monitor registration, appointments, check-in, token/queue and patient routing.
Hospital routing	Route OPD, walk-in, emergency, admission, diagnostics and discharge workflows.
Permissions	Control access by organization + location + department + action + data scope.

Replacement	Remove/suspend a receptionist and assign a replacement without changing patient ownership/history.
Performance	Monitor workload, wait time, registration volume and exceptions.
Communication	Hospital announcements and configured transactional communication/WhatsApp.

Role boundary

Receptionists remain operational staff. They do not diagnose, interpret reports, alter clinical notes, change clinical orders or make clinical emergency decisions. They route/escalate clinical issues to the appropriate team.

7. Patient & Treatment Management

Feature	Hospital Admin capability
Unified Patient ID	Search and use the existing Qlyno Patient ID instead of creating a second hospital identity.
Registration	Register patients who do not yet exist and flag potential duplicates.
Patient connection	Establish hospital relationship and permitted access to relevant history.
OPD	Registration, appointment, queue, consultation and follow-up coordination.
IPD	Admission, room/bed assignment, transfer, treatment progress and discharge workflow.
Patient timeline	View operational timeline according to permissions.
Documents	Manage hospital-generated documents, consent, discharge summaries and relevant records.
Billing	View billing status and coordinate financial workflows.
Consent	Respect patient/provider sharing and access controls.
Discharge	Coordinate discharge readiness, settlement, documents and follow-up.

8. Emergency / SOS Management

Emergency is a high-stakes workflow. The Qlyno patient journey explicitly identifies Emergency/SOS, ambulance, family alerts, hospital admission and emergency reliability as core product requirements.

Emergency principle

The system must optimize for fast routing and accountable human response, not autonomous clinical decision-making. Every alert needs a delivery state, owner, escalation path and audit trail.

8.1 Existing Hospital Patient — Emergency from Patient Profile

1. Patient opens the Qlyno app and presses the Emergency / SOS action.
2. Qlyno authenticates the patient and identifies the persistent Qlyno Patient ID.
3. System checks whether the patient has an active/current treatment relationship with a Qlyno-connected hospital.
4. If an active hospital relationship exists, the configured treating hospital becomes the primary emergency destination.
5. The hospital's Emergency workflow receives a high-priority alert containing permitted identity, location and relevant emergency context.
6. The hospital Emergency/Reception/Nurse/clinical team is notified according to its configured escalation policy.
7. If ambulance service is configured and an available ambulance exists, the system creates/initiates an ambulance dispatch workflow.
8. Hospital receives an expected-arrival/pre-arrival alert so the emergency team can prepare.
9. Patient/family receives delivery and status updates where configured.
10. If the primary hospital cannot accept the emergency or no configured ambulance/resource is available, Qlyno follows the hospital's fallback/escalation policy and may route to a nearby Qlyno-connected emergency hospital.
11. Every step is logged: SOS created, hospital notified, acknowledgement, ambulance status, fallback, arrival and closure.

8.2 Emergency Patient Not Currently Treated by a Hospital

12. Patient activates Emergency / SOS.
13. Qlyno obtains the patient's permitted location information.
14. System identifies configured nearby Qlyno-connected hospitals that advertise emergency capability and current availability signals.
15. The system presents/uses the configured emergency-routing policy to select a suitable destination.
16. Hospital receives the emergency alert and acknowledgement workflow.
17. Ambulance dispatch is initiated only where a participating ambulance service is available and the workflow supports dispatch.
18. If no hospital/ambulance acknowledges, escalation/fallback rules activate.
19. Patient/family receives status where available.

8.3 Emergency Admin Control Panel

Panel	What Admin can do
Live emergency board	See active SOS cases, status, location, destination, acknowledgement and owner.
Accept / acknowledge	Acknowledge administrative receipt and route to clinical emergency team.
Emergency capacity	Configure/report available emergency beds/resources where supported.
Ambulance	See available/busy/offline ambulances and active dispatches.
Escalation	Escalate unanswered alerts to configured roles.
Hospital fallback	Manage configured fallback/transfer routing rules.
Family alert	Trigger/monitor approved emergency family notifications.
Closure	Close emergency event only when the responsible workflow is complete/handed off.
Audit	Review the complete event trail and response times.

9. Ambulance Management

Feature	Requirement
Ambulance registry	Vehicle ID, type, equipment capability, base location and status.
Driver/team	Assign driver/crew and contact details according to hospital policy.
Availability	Available, dispatched, en route, at scene, transporting, at hospital, maintenance/offline.
Dispatch	Create dispatch linked to emergency case and patient where permitted.
Location	Track current/last-known location where the integration supports it.
ETA	Display estimated arrival when a reliable location/dispatch integration provides it.
Hospital destination	Primary hospital or configured receiving hospital.
Fallback	Escalate/re-route if ambulance or receiving facility cannot accept.
History	Maintain dispatch and transport records.

10. OT & Surgical Operations

The supplied HMS Hospital Admin foundation includes OT management. Qlyno's broader PRD adds surgical case readiness, blocker tracking, surgeon workflow, implant/equipment requirements and vendor coordination.

Feature	Hospital Admin capability
Surgical case	Create/manage surgery case linked to patient and care case.
Readiness checklist	Track pre-op assessment, investigations, consent, blood/implant/equipment and other configured dependencies.
OT scheduling	Allocate OT, date/time, team and resources.
Internal surgeon	Assign hospital-affiliated surgeon.
External surgeon request	Request an external/on-call surgeon when an internal surgeon is unavailable.
Surgeon response	Receive accept/decline/availability response and maintain request history.
Case blockers	Show missing dependency, owner and deadline.
Vendor dependency	Create procurement/vendor requests for implants, equipment, consumables or other requirements.
Post-op	Connect recovery, nursing tasks, documents and follow-up.

10.1 Surgeon Request Workflow

20. Hospital Admin/authorized clinical coordinator creates surgeon request.
21. Select specialty/sub-specialty, case type, required time, location and urgency.
22. Attach permitted case details and readiness information.
23. Send request to eligible surgeons/network.
24. Surgeon accepts, declines or requests clarification.
25. Hospital selects/assigns surgeon subject to authorization and credential checks.
26. Qlyno links surgeon to the surgical case with limited case-specific access.
27. Case readiness and surgeon availability are visible in the surgical control view.
28. After completion, surgeon affiliation/case access expires according to policy.

11. Vendor Request & Procurement Management

Feature	Requirement
Create request	Hospital can request equipment, implant, consumable, medicine, service or other configured product.
Case-linked request	Link procurement requirement to patient/surgical case when appropriate.
Vendor discovery	Use approved vendor network or manually selected vendor.
Quote collection	Receive and compare quotes, availability and delivery timelines.
Vendor review	Hospital Admin reviews vendor response, documents and commercial details.

Approval	Configured approval workflow for purchase/selection.
Purchase order	Generate/order approved item or service.
Delivery tracking	Track dispatched, delivered, rejected, delayed and received status.
Quality check	Record receiving/inspection outcome where required.
Audit	Keep complete request → quote → selection → order → delivery trail.

Important

Vendor requests should be structured workflow objects, not informal chat. The Qlyno product model already defines Procurement Request as a core object and expects hospital requirements, quotes, orders and delivery status to be traceable.

12. Diagnostics, Pharmacy, Billing & Insurance

Module	Hospital Admin capabilities
Laboratory	Investigations, sample workflow, reports, critical results, inventory and diagnostic coordination.
Radiology	X-ray/CT/MRI/Ultrasound requests, scheduling/status and report workflow.
Pharmacy	Medicine inventory, dispensing, purchase orders, stock levels and pharmacy operations.
Billing	Invoices, payments, refunds, outstanding, settlement and financial reports.
Insurance/TPA	Verification, pre-authorization, claims, approvals and settlement.
Inventory/Store	Medical supplies, consumables, procurement and stock control.
Assets	Equipment lifecycle, maintenance, depreciation and asset tracking.

13. Hospital & Doctor Verification

Manual verification is a trust and safety requirement for hospital and doctor public presence. The hospital should not become publicly searchable as a verified institution until required verification is completed.

Verification object	Suggested verification workflow
Hospital identity	Legal/registered name, address, contact details and organization ownership evidence.
Hospital registration	Upload applicable registration/license/certification documents.
Facility evidence	Proof of facility/operation and configured services where required.
Admin identity	Verify the person claiming hospital administration rights.
Doctor affiliation	Hospital confirms doctor affiliation; platform independently verifies professional documents as required.
Specialty/qualification	Review submitted professional documents before displaying verified claims.
Ambulance capability	Verify ambulance/service information before showing it as available capability.
Emergency capability	Hospital configures emergency availability and operating hours; public claims should reflect verified configuration.
Verification status	Pending → Under Review → Verified / Rejected / Needs More Information / Suspended.

13.1 Public Search Rules

- Verified hospital profile can appear in Qlyno hospital search.
- Doctors affiliated with the hospital can appear in doctor search only according to doctor verification/publication rules.
- Hospital affiliation must be visible on the doctor's profile without implying that the hospital endorses every claim beyond verified affiliation.
- Suspended/expired verification should remove or downgrade public verified status immediately.
- Private patient data, internal staffing data, bed-level details and emergency operational details must never appear on the public profile.

14. Security, Privacy & Access Control

Qlyno's product principles require role-specific simplicity, human-in-control clinical workflows, one source of truth and auditable sensitive actions. Hospital Admin should therefore have broad operational power without bypassing audit or clinical accountability.

Control	Requirement
RBAC + scope	Authorization = role + organization + branch/location + department + action + data scope.
MFA	Require multi-factor authentication for Hospital Admin and other privileged roles.
Session control	Device/session list, revoke session, inactivity timeout and suspicious-login alerts.
Sensitive action confirmation	Require step-up confirmation for high-risk actions such as disabling staff, changing permissions or emergency overrides.
Audit logs	Record actor, action, timestamp, entity, before/after and reason for sensitive actions.
Break-glass access	Emergency access may temporarily expand visibility only when policy allows; reason and duration are mandatory and audited.

Data minimization	Staff see only the patient information required for their role/task.
Consent	Respect patient sharing/consent rules when accessing cross-organization history.
Public/private separation	Public hospital/doctor profiles never expose private clinical or operational data.
Access revocation	Suspended/removed users lose active access; cached/secondary paths must not retain authorization.
Document security	Sensitive documents use permission checks and controlled download/share.
Backup/recovery	Critical hospital configuration and audit data require backup and recovery controls.

15. Hospital Admin — 'Can Do Everything' Model

The requested model is supported as an operational super-role inside the hospital, but it should be implemented as delegated capability rather than hidden privilege. When Admin performs another role's operational action, the system should record the Admin as the actor and the delegated role/context where applicable.

Capability	Admin access
Reception	Can perform registration, appointment, check-in, queue and routing actions.
Nurse Station	Can manage nurses, support staff, shifts, rosters, tasks and handovers.
Doctor operations	Can manage profiles, affiliation, schedules and operational tasks; cannot make/alter clinical decisions on behalf of a doctor without a licensed clinician acting.
Billing	Can view/manage permitted invoices, payments, refunds and financial settings.
Lab/Radiology	Can manage operational workflow and status; clinical interpretation remains with qualified professionals.
Pharmacy	Can manage inventory/operations; dispensing/clinical decisions remain with authorized pharmacy/clinical staff.
OT	Can schedule/coordinate; clinical surgical decisions remain with surgeon/clinician.
Emergency	Can coordinate/route/acknowledge/escalate; clinical triage remains with emergency clinical team.
Vendor	Can create/review/approve procurement requests according to approval rules.
Reports	Hospital-wide operational reporting subject to sensitive-data permissions.

Audit requirement

Never make Admin actions look as though they were performed by a doctor, nurse or receptionist. Example: 'Performed by Hospital Admin • acting within Reception workflow' rather than falsely attributing the action to a receptionist.

16. Notification & Escalation Center

Event	Recipients
New appointment/check-in	Relevant receptionist + doctor/team
Doctor delay	Admin + affected reception/department/patient workflow
Critical lab/result	Responsible clinician + relevant team
Emergency SOS	Emergency team + configured admin/coordination roles
Ambulance dispatch	Emergency team + dispatch/driver + patient/family where configured
Surgery blocker	Responsible surgeon + OT coordinator + Admin
Surgeon request	Eligible surgeon(s) + case owner
Vendor quote	Procurement/Admin reviewer
Vendor delivery delay	Request owner + department/case owner
Bed shortage	Admin + admissions/emergency team
Staffing gap	Nurse Station/Admin + department head
Security alert	Admin/security role

17. Reports & Analytics

Category	Reports
Patient flow	OPD volume, wait time, admission/discharge, transfers, emergency response.
Bed/ward	Occupancy, turnover, blocked beds, average stay.
Doctors	Appointments, utilization, wait time, cancellations, hospital affiliation.
Nursing	Staffing, attendance, workload, task completion and handover.
Reception	Registration volume, queue performance, no-shows and exceptions.
Surgery	OT utilization, case readiness, cancellations, blockers and turnaround.
Diagnostics	Pending tests, turnaround, critical reports and delays.
Pharmacy	Stock, dispensing, shortages, expiry and procurement.
Finance	Revenue, collection, outstanding, refunds, insurance/TPA.
Vendor	Requests, quote turnaround, purchase orders, delivery and delays.
Emergency	SOS count, acknowledgement time, dispatch status and resolution/transfer status.

Security	Login events, permission changes, break-glass use, audit anomalies.
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18. Global Search & Quick Actions

- Search patients by Qlyno Patient ID, name, phone or permitted identifiers.
- Search doctors by name, specialty, department, affiliation and verification status.
- Search staff by role, department, location and status.
- Search appointments, encounters, admissions, beds, wards, surgeries, reports, invoices and procurement requests.
- Search active emergency cases and their status subject to emergency permissions.
- Search vendors, quotes, purchase orders and delivery status.

Quick Action	Purpose
Add Doctor	Invite/create doctor and begin verification.
Add Staff	Create nurse, receptionist or support staff.
Create Department	Configure department and head.
Emergency	Open emergency control board.
Allocate Bed	View and allocate available bed.
Create Surgery Case	Start surgical workflow.
Request Surgeon	Create external/internal surgeon request.
Request Vendor	Create procurement/vendor requirement.
Dispatch Ambulance	Start ambulance workflow for eligible emergency case.
Generate Report	Open hospital analytics/report builder.

19. Core Hospital Workflows

19.1 Patient → Hospital → Treatment

29. Patient discovers hospital or doctor through Qlyno.
30. Patient books or arrives as walk-in/emergency.
31. Hospital identifies existing Qlyno Patient ID or creates/flags a new identity.
32. Reception completes permitted registration/check-in.
33. Patient enters OPD/IPD/emergency workflow.
34. Doctor/nurse/diagnostic/pharmacy teams execute their role-specific work.
35. Qlyno creates tasks and next actions from clinical/operational events.
36. Admission/bed/OT/diagnostic/pharmacy/billing workflows continue as needed.
37. Discharge creates documents, settlement and follow-up tasks.
38. Patient continues care through the unified Qlyno timeline.

19.2 Emergency SOS → Existing Treating Hospital

39. Patient presses SOS.
40. Qlyno identifies active hospital relationship.
41. Primary hospital receives emergency alert.
42. Emergency team acknowledges.
43. Hospital prepares/initiates pre-arrival workflow.
44. Ambulance dispatch is initiated if configured and available.
45. Patient/family receive status updates where enabled.
46. If no acknowledgement/acceptance, configured escalation/fallback runs.
47. Emergency event is linked to the patient's care case.
48. Event closes after handoff/resolution and retains audit trail.

19.3 Surgery → Surgeon + Vendor

49. Doctor/authorized clinician determines that surgery is required.
50. Surgical case is created.
51. Qlyno checks readiness dependencies.
52. If surgeon is unavailable, Hospital Admin creates surgeon request.
53. If equipment/implant/consumable is unavailable, Admin creates vendor request.
54. Surgeon and vendor responses are tracked.
55. Case readiness becomes visible as complete/missing/overdue.

- 56. OT is scheduled when required dependencies are ready.
- 57. Post-op nursing/diagnostic/discharge/follow-up tasks continue.

20. Recommended Core Data Model

Entity	Purpose	Important fields
Hospital	Organization	hospital_id, name, verification_status, locations, services, emergency_capability
HospitalLocation	Branch/campus	location_id, address, departments, operating_hours
Department	Operational unit	department_id, head, specialty, location, status
HospitalStaff	Common staff identity	staff_id, role, department, scope, status
DoctorAffiliation	Doctor ↔ hospital	doctor_id, hospital_id, specialty, verification, start/end
NurseStation	Nursing unit	station_id, department/location, lead, status
EmergencyEvent	SOS workflow	event_id, patient_id, location, hospital, priority, status, timestamps
AmbulanceDispatch	Emergency transport	dispatch_id, ambulance, driver, case, origin, destination, status
SurgicalCase	Surgery workflow	case_id, patient, surgeon, OT, readiness, blockers, status
SurgeonRequest	Specialist request	request_id, case_id, specialty, urgency, recipients, response
ProcurementRequest	Vendor requirement	request_id, case_id, product/service, quotes, approval, delivery
VendorQuote	Vendor response	quote_id, vendor, price, availability, SLA, documents
Bed	Bed resource	bed_id, ward, status, patient, isolation/capability
AuditEvent	Security trace	actor, role_context, action, entity, before_after, timestamp, reason
VerificationCase	Hospital/doctor verification	subject, documents, reviewer, status, decision, reason

21. Acceptance Criteria

- Hospital Admin can manage hospital departments, doctors, nurses, receptionists and support staff.
- Hospital can have multiple receptionists with location/department/desk scope.
- Hospital can create and manage Nurse Stations and multiple nurses.
- Hospital can configure shifts, attendance and workforce allocation.
- Hospital Admin can perform operational actions available to other staff roles where permission allows.
- Admin actions remain attributable to the Admin and are never falsely attributed to another staff member.
- Hospital Admin can manage OPD, IPD, wards, beds, emergency and OT workflows.
- Existing Qlyno Patient ID is reused across hospital and other providers.
- Patient emergency SOS can be linked to an active treating hospital relationship.
- Emergency alert has acknowledgement/delivery status and escalation path.
- Ambulance dispatch is attempted only when an eligible configured resource is available.
- Emergency fallback can route toward a configured nearby Qlyno-connected hospital when primary handling is unavailable.
- Hospital can request internal or external surgeons and track responses.
- Hospital can create vendor/procurement requests and review quotes.
- Surgical cases show readiness blockers, owner and deadline.
- Hospital and doctor verification supports manual review and status lifecycle.
- Verified doctors affiliated with the hospital can appear in Qlyno search subject to publication rules.
- Suspended/rejected verification removes or limits verified public status.
- Hospital Admin access is protected by RBAC, scope checks, MFA and audit logs.
- Emergency break-glass access, if enabled, is time-limited and audited.
- Sensitive patient data is never exposed merely because a user is a Hospital Admin.
- Clinical decisions remain attributable to licensed clinicians.
- All sensitive actions are auditable.

22. Additional High-Value / Differentiating Features

The following are product proposals added on top of the supplied HMS/Qlyno foundation. They are not presented as existing source requirements.

Proposed feature	Why it is useful
Hospital Command Center	One screen for active emergencies, bed pressure, staffing gaps, OT blockers and overdue operational tasks.
Emergency Pre-Arrival Packet	When a known patient triggers SOS, send a permitted, compact patient/care summary to the receiving emergency team before arrival.
Emergency Capacity Signal	Hospital can publish configured emergency/bed capability signals to Qlyno so routing decisions are based on more than distance.
Automatic Escalation Ladder	If emergency alert is not acknowledged within configured time, escalate from reception → emergency coordinator → clinical team → admin.

Care Continuity Card	At emergency arrival, show the hospital's active treatment relationship, allergies/critical information only where authorized, and current care case.
Operational Break-Glass	Controlled temporary access for exceptional emergencies with mandatory reason, time limit and audit.
Case Readiness Score	Surgical cases get a readiness view showing completed, missing and overdue dependencies.
Staff Coverage Heatmap	Visualize staffing shortages by department, shift and skill.
Hospital Service Availability	Public profile can show configured services/hours such as emergency, lab, pharmacy, ambulance and specialties.
Verification Expiry Alerts	Notify Admin before hospital/doctor documents or certifications expire.
Vendor Reliability Score	Track vendor delivery timeliness, rejection rate and fulfillment history internally.
Surgeon Reliability View	Track response time and accepted/completed case history for network surgeons where permitted.
Bed Turnaround Task	When a bed becomes vacant, automatically create cleaning/inspection/readiness tasks before marking it available.
Incident Management	Create and track operational incidents such as equipment failure, staffing shortage, security event or patient-flow blockage.
Admin Delegation	Admin can temporarily delegate selected capabilities to a trusted manager without sharing credentials.
Hospital Health Score	Operational score based on staffing, beds, emergency response, pending tasks, OT blockers and compliance exceptions.

23. Recommended Admin Permission Matrix

Module	View	Create/Edit	Approve/Override	Clinical decision
Departments	Full	Full	Full	No
Doctors	Full permitted	Profile/affiliation	Verification workflow	No
Nurses/Staff	Full operational	Full	Full within policy	No
Reception	Full operational	Full	Full	No
Patients	Permitted hospital scope	Operational	Break-glass where allowed	No
Emergency	Full emergency ops	Full operational	Emergency routing/escalation	No
Beds/Wards	Full	Full	Allocation override	No
OT/Surgery	Operational	Schedule/resources	Operational override	No
Billing	Full finance scope	Full	Approval per policy	No
Vendor	Full procurement scope	Full	Approve per policy	No
Reports	Hospital scope	Generate	N/A	No
Security	Full audit	Permissions	Security override with audit	No

24. Recommended Hospital Admin Navigation

Navigation	Sub-sections
Dashboard	Hospital today, command center, alerts, blockers
Departments	Departments, heads, locations, services
Doctors	Doctors, affiliations, schedules, verification
Nurse Station	Stations, nurses, support staff, shifts, rosters
Reception	Receptionists, desks, queues, front-desk performance
Patients	Patient search, admissions, treatment, documents
Appointments / OPD	Appointments, queues, walk-ins
IPD	Admissions, transfers, discharge
Wards & Beds	Wards, beds, occupancy, readiness
Emergency	SOS, active cases, ambulance, escalation
OT & Surgery	Cases, readiness, OT schedule, surgeon requests
Diagnostics	Lab, radiology, reports, critical results
Pharmacy	Inventory, dispensing, purchase
Billing & Insurance	Invoices, payments, refunds, TPA
Inventory & Assets	Stores, assets, maintenance
Vendors	Requests, quotes, orders, delivery
Reports	Operational, financial, clinical-operations, workforce
Quality & Compliance	Policies, audits, verification, incidents
Security & Audit	Users, permissions, sessions, logs, break-glass
Settings	Hospital profile, notifications, integrations, emergency rules
Global Search	Patients, doctors, staff, cases, requests, records

25. Source Alignment & Boundary

This PRD is grounded in the uploaded HMS Hospital Admin documentation, the Qlyno Product Requirements Document, the Qlyno Patient Journey, the Receptionist module and the Nurse Station module.

- The supplied HMS Hospital Admin module defines the core hospital administration foundation: departments, doctors, nurses/staff, reception, patients, appointments, OPD/IPD, wards, beds, emergency and OT. 穀filecite學turn4file3係
- The broader HMS foundation includes pharmacy, laboratory, radiology, billing, insurance/TPA, inventory, assets, ambulance, shifts, attendance, notifications, reports, audit logs, documents, communication, quality/compliance, settings, role/permissions and global search. 穀filecite學turn3file0係
- Qlyno's product architecture defines Hospital Admin as responsible for departments, staff, beds, services, finance, operations and performance, while also connecting surgical operations and the vendor network. 穀filecite學turn4file5係
- The Qlyno object model defines Care Case, Surgical Case, Procurement Request, Task, Notification and Timeline Event as cross-role workflow objects. 穀filecite學turn4file5係
- The Qlyno Patient Journey explicitly includes Emergency/SOS, ambulance/family alerts, hospital admission, bed/ward management, nursing, surgeon coordination and surgery workflow. 穀filecite學turn4file18係
- The Patient module requires one persistent Qlyno Patient ID, permission-based provider access and auditable emergency escalation. 穀filecite學turn3file13係 穀filecite學turn3file18係
- The Receptionist module defines hospital reception, emergency routing and the context + role + scope permission model. 穀filecite學turn4file1係
- The Nurse Station module defines hospital-wide admin oversight, nursing stations, nurses, support staff, rosters, handovers and auditable assignments. 穀filecite學turn4file2係

Proposed additions

Hospital manual verification, the exact treating-hospital SOS routing logic, emergency capacity signals, pre-arrival packet, automated escalation ladder, admin delegation, vendor/surgeon reliability views and the Hospital Health Score are product proposals built from the user's requested direction and the existing Qlyno architecture. They should be validated before engineering commitment.

26. Product Principle

The hospital should operate as one connected system.

A patient should not have to repeat their identity or manually explain where they are being treated. A hospital should not have to chase a surgeon, vendor, nurse, receptionist or emergency response through disconnected channels. Qlyno should turn every important hospital event into a visible state, a next action, an owner, a deadline and an escalation path.